Examiner only

6. The following equation shows a fusion reaction that happens in stars.

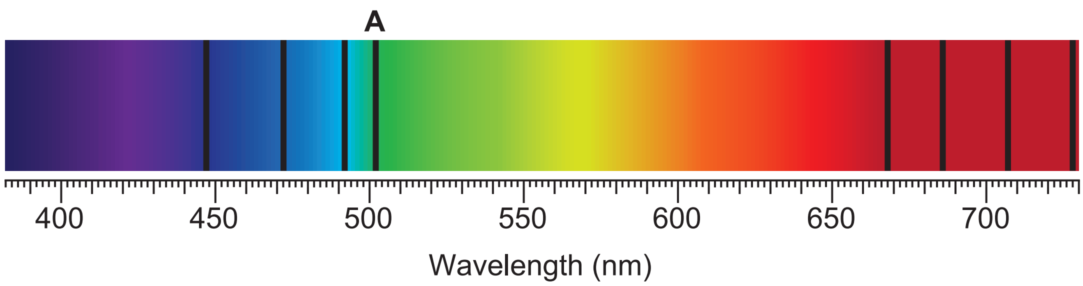
$$^{1}_{1}\mathrm{H} + {}^{2}_{1}\mathrm{H} \longrightarrow {}^{3}_{2}\mathrm{He} + \text{energy}$$

(a) State the **two** conditions needed for a fusion reaction to take place in a star. [2]

1. ....

2. ....

(b) An incomplete absorption spectrum from the Sun is shown below.



(i) An absorption line at wavelength **590** × **10^{-9} m** is missing from the spectrum. **Complete the diagram** above by adding the missing absorption line. [1]

(ii) Describe **and** explain how the position of the absorption line labelled **A** would be different for a star in a more distant galaxy. [3]

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